

Unit 2 Guided Notes — Exponents & Scientific Notation

Same base? Move the exponents. Big or tiny number? Pack it into a $\times 10^n$.

Standards: NC.8.EE.1 · NC.8.EE.3 · NC.8.EE.4

This packet weaves three parts together: **Guided Notes** with worked examples, **Practice** built into each section to try as you learn, and a **Cumulative Review** with a full **Answer Key** at the end. Fill in the blanks and show your work on your printed copy.

Topics Covered: 2.1 Exponent Rules — Product, Quotient, Power, Zero, Negative (NC.8.EE.1) · 2.2 Scientific Notation — Format & Conversions (NC.8.EE.3) · 2.2 Multiplying & Dividing Scientific Notation (NC.8.EE.4) · 2.2 Word Problems with Scientific Notation (NC.8.EE.4)

1 Exponent Rules

➤ **KEY RULE:** Exponent rules only apply when the BASES are the SAME!

Fill the examples and key phrases on your printed copy.

Rule	Formula	Simple Example	Monomial Example	Key Phrase
Product Rule	$x^a \times x^b = x^{a+b}$	_____	_____	_____
Quotient Rule	$x^a \div x^b = x^{a-b}$	_____	_____	_____
Power Rule	$(x^a)^b = x^{a \times b}$	_____	_____	_____
Zero Exponent	$x^0 = 1$	_____	_____	_____
Negative Exponent	$x^{-n} = 1/x^n$	_____	_____	_____

⚠ **PARENTHESES MATTER!**

$-3^2 = \underline{\hspace{2cm}}$ vs. $(-3)^2 = \underline{\hspace{2cm}}$

Without (), ONLY the base gets the exponent — not the negative sign!

PRODUCT RULE: COUNT THE FACTORS

$$\begin{array}{c} \boxed{x} \boxed{x} \boxed{x} \\ x^3 \text{ (3 factors)} \end{array} \times \begin{array}{c} \boxed{x} \boxed{x} \\ x^2 \text{ (2)} \end{array} = \begin{array}{c} \boxed{x} \boxed{x} \boxed{x} \boxed{x} \boxed{x} \\ x^5 \text{ (5 factors)} \end{array}$$

$$x^3 \times x^2 = x^{3+2} = x^5 \rightarrow \text{ADD the exponents}$$

Modeling the Rules — See WHY

Each rule comes from just writing out the repeated multiplication — **first base**, **second base**, result.

Product Rule · $x^3 \times x^2 = x^5$

See why → $\boxed{x} \boxed{x} \boxed{x} \times \boxed{x} \boxed{x} = \boxed{x} \boxed{x} \boxed{x} \boxed{x} \boxed{x} = \boxed{x^5}$

Count the factors: $3 + 2 = 5$. Same base → ADD the exponents.

Quotient Rule · $x^5 \div x^2 = x^3$

See why → $x \cdot x \cdot x \cdot x \cdot x \div x \cdot x \rightarrow$ cancel 2 pairs → $x \cdot x \cdot x = x^3$

Cancel matching factors: $5 - 2 = 3$. Same base → SUBTRACT the exponents.

Power Rule · $(x^2)^3 = x^6$

See why → $x^2 \cdot x^2 \cdot x^2 = x \cdot x \cdot x \cdot x \cdot x \cdot x = x^6$

Three copies of x^2 : $2 \times 3 = 6$. Power to a power → MULTIPLY.

Zero Exponent · $x^0 = 1$

See why → $x \cdot x \cdot x \div x \cdot x \cdot x = 1$, but by the quotient rule = $x^{3-3} = x^0$. So $x^0 = 1$

Anything (except 0) divided by itself is 1 → $x^0 = 1$.

Negative Exponent · $x^{-3} = 1/x^3$

See why → $x \cdot x \div x \cdot x \cdot x \cdot x \cdot x = x^{2-5} = x^{-3}$; cancelling leaves $1 / x^3$

A negative exponent means 'flip it' → $x^{-n} = 1/x^n$.

Quick Check

MC Simplify $x^5 \cdot x^3$. Which rule applies, and what is the result?

- ☐ Product Rule → x^8 ☐ Power Rule → x^{15} ☐ Quotient Rule → x^2

Match the Rule to its Key Phrase

MATCH Draw a line from each rule to its key phrase.

Rule

Key Phrase

Product Rule

Power to power? MULTIPLY

Quotient Rule

Same base? ADD exponents

Power Rule

Flip to a fraction

Negative Exponent

Same base? SUBTRACT exponents

Worked Example — Simplify $(4x^3y^2)(3x^5y)$

Step 1 — Multiply coefficients: $4 \times 3 = 12$

Step 2 — Add x exponents: $x^3 \cdot x^5 = x^8$

Step 3 — Add y exponents: $y^2 \cdot y^1 = y^3$

✓ **Step 4 — Combine:** $12x^8y^3$

ERROR ANALYSIS Spot the Error — Simplify $(3x^2)^3$. A student wrote $3x^6$.

The coefficient 3 must ALSO be cubed: $3^3 = 27$. Correct: $27x^6$. The power rule applies to EVERYTHING inside the parentheses.

PRACTICE A Exponent Rules with Monomials

Rules: Multiply → ADD exponents | Divide → SUBTRACT | Power-to-Power → MULTIPLY

- | | |
|------------------------------------|--------------------------------------|
| 1 Simplify $x^3 \cdot x^5$ _____ | 2 Simplify $(y^4)^3$ _____ |
| 3 Simplify $a^8 \div a^3$ _____ | 4 Simplify $(2x^3)(5x^2)$ _____ |
| 5 Simplify $(3m^2)^3$ _____ | 6 Simplify $12y^7 \div 4y^2$ _____ |
| 7 Simplify $(-2a^4)^2$ _____ | 8 Simplify $(5x^2y)(3xy^3)$ _____ |
| 9 Simplify $18n^6 \div 6n^6$ _____ | 10 Simplify $(4b^3)^2 \div 2b$ _____ |
| 11 Simplify $(-3x^2)(-4x^3)$ _____ | 12 Simplify $(x^5y^2)^3$ _____ |

2 Scientific Notation

$$a \times 10^n \quad \text{where} \quad 1 \leq a < 10$$

4.5×10^7 | coefficient · base (always 10) · exponent

SCIENTIFIC NOTATION: MOVE THE DECIMAL

$$45,000 \rightarrow 4.5 \times 10^4$$

big number → move LEFT 4 places → exponent +4

$$0.0067 \rightarrow 6.7 \times 10^{-3}$$

small number → move RIGHT 3 places → exponent -3

Type	Direction	Exponent	Example
Large number	Move decimal LEFT	_____	$45,000 = 4.5 \times 10^4$
Small number (< 1)	Move decimal RIGHT	_____	$0.0067 = 6.7 \times 10^{-3}$

PRACTICE B Scientific Notation Conversions

Remember: $a \times 10^n$ where $1 \leq a < 10$. Positive exp = BIG number · Negative exp = SMALL number.

- | | |
|---------------------------------------|--|
| 1 45,000,000 → sci. notation _____ | 2 0.00032 → sci. notation _____ |
| 3 6.02×10^8 → standard _____ | 4 9.1×10^{-5} → standard _____ |
| 5 780,000 → sci. notation _____ | 6 0.0000067 → sci. notation _____ |
| 7 3.45×10^4 → standard _____ | 8 1.2×10^{-3} → standard _____ |
| 9 5,280,000,000 → sci. notation _____ | 10 2.5×10^{-2} → standard _____ |

Multiplying Scientific Notation — 3-Step Process

1. _____ the coefficients (the a-values)
2. _____ the exponents
3. Adjust if coefficient is NOT between _____ and _____

Example (no adjustment): $(3 \times 10^4) \times (2 \times 10^5)$

Multiply coefficients: $3 \times 2 = 6$

ADD exponents: $4 + 5 = 9$

✓ **Answer:** 6×10^9

Dividing Scientific Notation — 3-Step Process

1. _____ the coefficients
2. _____ the exponents
3. Adjust if needed

Example (no adjustment): $(8 \times 10^6) \div (4 \times 10^2)$

Divide coefficients: $8 \div 4 = 2$

SUBTRACT exponents: $6 - 2 = 4$

✓ **Answer:** 2×10^4

⚠ ADJUSTMENT RULES:

- If coefficient ≥ 10 : Move decimal LEFT, ADD 1 to exponent
- If coefficient < 1 : Move decimal RIGHT, SUBTRACT 1 from exponent

Example needing adjustment: $(5 \times 10^3) \times (4 \times 10^2)$

Multiply: $5 \times 4 = 20$

Add exponents: $3 + 2 = 5 \rightarrow$ result: 20×10^5

Adjust ($20 \geq 10$): $20 = 2.0 \times 10^1$, so ADD 1 to exponent

✓ **Final answer:** 2.0×10^6

ERROR ANALYSIS **Spot the Error — Convert 0.0045 to scientific notation.** A student wrote 4.5×10^3 .

Small number $\rightarrow$ NEGATIVE exponent. Correct: 4.5×10^{-3} . Numbers less than 1 always have negative exponents in scientific notation.

PRACTICE C Operations with Scientific Notation

Steps: Multiply/divide the coefficients · Add/subtract the exponents · Adjust if the coefficient is not between 1 and 10.

- | | |
|---|---|
| 1 $(3 \times 10^4)(2 \times 10^5)$ _____ | 2 $(8 \times 10^7) \div (4 \times 10^3)$ _____ |
| 3 $(5 \times 10^3)(7 \times 10^6)$ _____ | 4 $(9.6 \times 10^8) \div (3.2 \times 10^5)$ _____ |

Word Problems with Scientific Notation

Strategy: Identify what operation to use. Write your scientific-notation setup. Adjust if needed.

PRACTICE D Word Problems

1. The Sun is about 9.3×10^7 miles from Earth. Light travels 1.86×10^5 miles per second. How many seconds does it take light to reach Earth?

Answer: _____

2. A bacteria cell is about 2×10^{-6} m long. A virus is about 5×10^{-8} m long. How many times longer is the bacteria?

Answer: _____

3. The US national debt is about 3.4×10^{13} dollars. The US population is about 3.4×10^8 people. What is the debt per person?

Answer: _____

4. A grain of sand has mass 6.5×10^{-4} grams. A beach has about 8×10^{18} grains of sand. What is the total mass in grams?

Answer: _____

5. There are about 3.7×10^{13} cells in a human body. Each cell is about 1×10^{-5} m wide. If all cells were lined up, how long in meters?

Answer: _____

6. A computer does 4.8×10^9 calculations per second. A task needs 1.44×10^{12} calculations. How many seconds to finish?

Answer: _____

★ Cumulative Practice — Unit 2 Review ★

Mixed practice over everything above. Show all work; circle your final answer.

REVIEW · A Exponent Rules

- 1 Simplify $x^4 \cdot x^6$ _____
- 2 Simplify $(m^3)^4$ _____
- 3 Simplify $p^9 \div p^4$ _____
- 4 Simplify $(3a^2)(4a^5)$ _____
- 5 Simplify $(-2c^4)^3$ _____
- 6 Simplify $(6x^5y^3) \div (2x^2y)$ _____

REVIEW · B Scientific Notation Conversions

- 1 $6,300,000 \rightarrow$ sci. notation _____
- 2 $0.00071 \rightarrow$ sci. notation _____
- 3 $2.08 \times 10^5 \rightarrow$ standard _____
- 4 $4.5 \times 10^{-6} \rightarrow$ standard _____
- 5 $91,000,000 \rightarrow$ sci. notation _____
- 6 $3.6 \times 10^{-2} \rightarrow$ standard _____

REVIEW · C Operations

- 1 $(2 \times 10^3)(4 \times 10^6)$ _____
- 2 $(9 \times 10^8) \div (3 \times 10^2)$ _____
- 3 $(6 \times 10^5)(5 \times 10^4)$ _____
- 4 $(7.2 \times 10^7) \div (1.8 \times 10^3)$ _____

REVIEW · D Word Problems

1. A red blood cell is about 8×10^{-6} m wide. A platelet is about 2×10^{-6} m wide. How many times wider is the red blood cell?

Answer: _____

2. Earth is about 1.5×10^8 km from the Sun. A spacecraft travels 3×10^4 km per hour. How many hours would the trip take?

Answer: _____

3. A factory makes 5×10^3 widgets per day. How many widgets does it make in 2×10^2 days?

Answer: _____